

SARATOV TSENTRALNY, Russian Federation

WMO: 341720

Lat: 51.5659N

Lon: 46.0389E

Elev: 499

StdP: 14.43

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.9	(c) -5.3	(d) -15.4	(e) 2.4	(f) -8.9	(g) -10.8	(h) 3.1	(i) -4.1	(j) 26.3	(k) 21.5	(l) 23.2	(m) 21.5	(n) 8.6	(o) 310	(p) 0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.4	(c) 92.3	(d) 67.7	(e) 88.9	(f) 67.2	(g) 85.6	(h) 66.3	(i) 70.5	(j) 85.0	(k) 69.2	(l) 83.3	(m) 67.8	(n) 81.4	(o) 9.2	(p) 100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	98.3	75.1	64.5	92.6	74.1	62.9	87.3	73.6	34.7	85.2	33.6	83.2	32.5	81.0	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	19.3	17.0	DB	-14.4	96.4	6.4	3.7	-19.0	99.1	-22.8	101.3	-26.4	103.4	-31.1	106.1
			WB	-15.0	74.1	6.4	1.9	-19.6	75.4	-23.4	76.5	-27.0	77.6	-31.6	78.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.2	16.7	17.3	28.2	47.3	61.3	68.8	72.8	70.7	59.3	45.1	31.4	21.1		
	DBStd	22.53	11.50	11.62	9.23	9.67	8.76	7.80	6.51	7.60	8.16	8.24	9.42	10.93		
	HDD50	4467	1032	916	675	163	13	1	0	0	15	198	557	897		
	HDD65	7970	1497	1336	1139	534	179	48	10	32	208	618	1007	1362		
	CDD50	2698	0	0	0	83	362	566	708	640	294	45	0	0		
	CDD65	727	0	0	0	3	63	163	254	207	37	0	0	0		
	CDH74	6363	0	0	0	20	492	1434	2297	1870	250	0	0	0		
	CDH80	2322	0	0	0	1	120	512	894	742	53	0	0	0		
(14) Wind	WSAvg	9.6	10.1	10.2	10.3	9.8	9.4	9.0	8.5	8.6	8.9	9.6	10.2	10.4		
(15) Precipitation	PrecAvg	17.0	1.5	1.1	1.1	0.9	1.5	1.8	1.6	1.7	1.5	1.3	1.7	1.5		
	PrecMax	24.8	3.4	3.0	3.1	2.0	3.5	5.6	4.9	4.6	4.4	3.5	4.1	4.6		
	PrecMin	10.0	0.1	0.0	0.0	0.2	0.0	0.4	0.3	0.0	0.0	0.0	0.2	0.3		
	PrecStd	3.7	0.7	0.7	0.7	0.6	1.0	1.3	1.0	1.1	1.1	0.9	0.9	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.5	37.8	55.0	77.0	87.7	95.2	96.9	97.0	85.6	69.0	53.7	42.6		
		MCWB	37.2	35.6	46.3	57.8	63.6	67.3	68.3	68.0	62.2	56.7	50.5	41.8		
	2%	DB	35.7	35.8	47.7	71.7	83.5	89.7	91.8	91.7	80.5	64.4	49.5	37.8		
		MCWB	34.5	34.1	41.7	54.8	63.0	67.4	68.1	67.7	61.5	54.0	46.9	36.5		
	5%	DB	33.8	34.1	42.9	66.6	79.4	85.8	88.2	87.8	76.6	60.5	46.3	35.8		
		MCWB	32.7	32.8	38.9	52.8	61.5	66.6	67.6	66.7	59.8	52.4	44.7	34.7		
	10%	DB	32.1	32.4	39.2	62.5	75.5	82.2	84.6	84.0	72.0	56.9	43.0	34.0		
		MCWB	31.3	31.4	36.0	50.2	59.8	65.3	66.6	65.5	58.2	50.5	41.5	33.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.4	36.4	46.7	60.0	67.4	72.8	72.4	71.6	66.5	60.1	51.4	41.4		
		MCDB	37.9	37.0	53.1	72.3	81.6	84.8	87.4	86.9	76.7	65.2	52.7	41.9		
	2%	WB	34.4	34.4	42.2	56.4	64.8	70.1	70.6	69.7	63.5	56.0	47.1	36.9		
		MCDB	34.9	35.2	46.1	68.0	79.3	83.3	85.6	84.8	74.0	61.0	48.4	37.6		
	5%	WB	32.6	33.0	39.1	53.9	62.9	68.3	69.3	68.2	61.5	53.5	44.4	34.9		
		MCDB	33.3	33.9	42.3	64.8	76.6	81.3	83.6	83.2	72.7	59.0	45.6	35.6		
	10%	WB	31.3	31.6	36.3	51.3	60.9	66.7	68.0	66.9	59.6	51.1	41.4	33.1		
		MCDB	32.1	32.5	38.7	60.7	72.8	79.2	81.2	80.6	70.4	55.7	42.5	33.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	9.9	10.5	15.6	17.1	17.0	17.4	17.8	15.6	11.9	8.2	8.4		
		MCDBR	7.9	8.1	14.0	21.5	20.8	21.3	21.6	22.1	20.8	17.2	10.5	7.4		
	5% WB	MCWBR	7.8	8.0	10.1	10.9	10.0	9.0	8.1	8.1	9.5	10.3	9.0	7.2		
		MCWBR	7.8	8.0	12.8	19.5	19.5	18.9	18.6	19.5	18.0	14.4	9.9	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.299	0.331	0.365	0.429	0.400	0.387	0.404	0.422	0.376	0.350	0.327	0.301		
		taud	2.113	2.033	2.084	2.109	2.302	2.376	2.329	2.262	2.366	2.376	2.286	2.148		
	Ebn at Noon		215	240	258	255	271	275	268	255	256	239	208	193		
		Edh at Noon	24	35	41	45	39	36	38	38	31	25	21	20		
(44) All-Sky Solar Radiation	RadAvg		303	650	1073	1405	1834	1982	1952	1651	1122	594	259	178		
	RadStd		68	106	178	123	161	168	153	127	142	89	44	49		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+335	+200
(47) Regional (5 neighbors)		+1.24	N/A	N/A	+2.25	N/A	N/A	N/A	+322	+139

Nomenclature: See separate page